

How to give read and write permissions using Unix commands.
A very simple example is to create a text file with a C program,
then use chmod.

Create a file called permission.c

```
#include <stdio.h>

int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

Compile

gcc permission.c -o permission

Run

Your text here 1

./permission

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read

Others have: Read

This is permission 644.

Give Read and Write permission to everyone

```
chmod 666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

`chmod u+rw sample.txt`

`chmod o+rw sample.txt`

`chmod a+rw sample.txt`

Task of the Day

- 1) Give Read and Write only to the owner - **chmod 600 sample.txt**
- 2) Give Read and Write to the owner and group - **chmod 660 sample.txt**
- 3) Only owner can read and write - **chmod 600 sample.txt**
- 4) Owner read/write, everyone else read-only - **chmod 644 sample.txt**
- 5) Owner can do everything, others can only read and execute - **chmod 705 sample.txt**
- 6) Owner and group can read/write - **chmod 660 sample.txt**
- 7) Only owner can execute - **chmod 100 sample.txt**

Fill in the permission numbers for the following:

- 8) Owner: read + write; Group: nothing; Others: nothing
→ 600
- 9) Owner: read + write; Group: read; Others: read →
- 10) Owner: read + write + execute; Group: read + execute;
Others: read + execute →
- 11) Owner: read + write; Group: read + write; Others:
nothing →
- 12) Everyone: read only →

```
pavan_kumar@DESKTOP-GQI43H3:~$ cd 2520090038_OSLab
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab$ cd Class
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ nano Permission.c
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ gcc Permission.c -o Permission
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ./Permission
File created successfully.
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l Permission
-rwxr-xr-x 1 pavan_kumar pavan_kumar 16096 Aug 17 05:55 Permission
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 600 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
-rw----- 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 660 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
-rw-rw---- 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 600 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
-rw----- 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 622 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
-rw--w--w- 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 644 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
-rw-r--r-- 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 755 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
-rwxr-xr-x 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 660 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
-rw-rw---- 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 100 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
---x----- 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ chmod 705 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ ls -l sample.txt
-rwx---r-x 1 pavan_kumar pavan_kumar 28 Aug 17 05:55 sample.txt
pavan_kumar@DESKTOP-GQI43H3:~/2520090038_OSLab/Class$ |
```